

37.2 Offsets and rates

A count on its own is rarely interesting. Twelve accidents mean one thing on a road carrying a thousand vehicles a day and another on one carrying a hundred thousand. What is comparable is the **rate**: the count divided by the exposure that produced it. The Poisson log-linear model handles rates without new machinery, by moving a known quantity into the linear predictor and refusing to give it a coefficient.

37.2.1 Exposure

Let $t_i > 0$ be the **exposure** of observation i : person-years of follow-up, vehicle-kilometres, population size, area surveyed. The natural assumption is that the expected count is proportional to exposure,

$$\mu_i = t_i \lambda_i,$$

where λ_i is the expected number of events per unit of exposure, the quantity of interest. Modelling λ_i log-linearly, $\log \lambda_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$, gives

$$\log \mu_i = \log t_i + \mathbf{x}_{(i)}^\top \boldsymbol{\beta}, \quad \mu_i = t_i \exp(\mathbf{x}_{(i)}^\top \boldsymbol{\beta}). \quad (37.2.1)$$

The term $\log t_i$ is an **offset**: a column of the linear predictor whose coefficient is known to be one rather than estimated. Coefficients are now log rate ratios, e^{β_j} multiplying the rate per unit of exposure rather than the raw count. This is *not* a regression of the observed rates y_i/t_i , which would throw away the information that a rate based on a thousand person-years is better determined than one based on five: the response stays the count and the exposure enters as a known constant.

Proposition 37.2.1. Offsets

Consider the model Eq. 37.2.1 with $\text{rank}(\mathbf{X}) = p$, and write $\boldsymbol{\mu} = \boldsymbol{\mu}(\boldsymbol{\beta})$ for the mean vector it defines. Then:

- (a) the log-likelihood is $\ell(\boldsymbol{\beta}) = \sum_i \{y_i(\log t_i + \mathbf{x}_{(i)}^\top \boldsymbol{\beta}) - t_i e^{\mathbf{x}_{(i)}^\top \boldsymbol{\beta}}\} - \sum_i \log y_i!$; its score is $\mathbf{X}^\top(\mathbf{y} - \boldsymbol{\mu})$ and its information is $\mathbf{X}^\top \mathbf{W} \mathbf{X}$ with $\mathbf{W} = \text{diag}(\mu_1, \dots, \mu_n)$, so Theorem 37.1.1 holds verbatim with the offset in place;
- (b) consequently Theorem 37.1.2 holds verbatim: the fitted counts, not the fitted rates, reproduce the observed margins of the model matrix, and with an intercept $\sum_i \hat{\mu}_i = \sum_i y_i$;
- (c) (**grouping**) suppose the rows of $\mathbf{X}$ take only G distinct values $\mathbf{x}_{(g)}$, with n_g rows equal to the g th, all with exposure $t_i = 1$. Let s_g be the total count of group g . Then fitting Eq. 37.1.1 to the n individual records and fitting Eq. 37.2.1 to the G totals s_g with exposures $t_g = n_g$ give the same $\hat{\boldsymbol{\beta}}$, the same information matrix and the same deviance differences between nested models;
- (d) (**a free exposure coefficient**) the model $\log \mu_i = \psi \log t_i + \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$ contains Eq. 37.2.1 as the submodel $\psi = 1$. Its fit coincides with the offset fit if and only if $\hat{\psi} = 1$, and the likelihood ratio statistic for $H_0 : \psi = 1$ is asymptotically $\chi^2(1)$.

Proof.

- (a) Substituting $\mu_i = t_i e^{\mathbf{x}_{(i)}^\top \boldsymbol{\beta}}$ into the Poisson log-likelihood gives the stated expression, and $\partial \mu_i / \partial \boldsymbol{\beta} = \mu_i \mathbf{x}_{(i)}$ exactly as before, because the factor t_i is a constant. The derivative of $\sum_i y_i \log t_i$ in $\boldsymbol{\beta}$ is zero. Everything in the proof of Theorem 37.1.1 then goes through with the new μ_i , which is (a); and (b) follows from (a) as before.
- (b) Within group g all the means equal $\mu_g = e^{\mathbf{x}_{(g)}^\top \boldsymbol{\beta}}$, so

$$\ell_{\text{ind}}(\boldsymbol{\beta}) = \sum_{g=1}^G \{s_g \log \mu_g - n_g \mu_g\} - \sum_i \log y_i!.$$

The grouped model has means $n_g \mu_g$, so its log-likelihood is

$$\ell_{\text{grp}}(\boldsymbol{\beta}) = \sum_{g=1}^G \{s_g (\log n_g + \log \mu_g) - n_g \mu_g\} - \sum_g \log s_g!.$$

The two differ by a quantity free of $\boldsymbol{\beta}$. Log-likelihoods differing by a constant have the same maximizer and the same derivatives of every order, hence the same information; and differences of deviances are differences of log-likelihoods, so the constant cancels there too.

- (c) Setting $\psi = 1$ recovers [Eq. 37.2.1](#), so the models are nested with one extra parameter. The larger model's log-likelihood is strictly concave in $(\boldsymbol{\beta}^\top, \psi)^\top$ whenever $[\mathbf{X}, \log \mathbf{t}]$ has full column rank, by Theorem 37.1.1(c) applied to that model matrix, so its maximizer is unique; it therefore agrees with the offset fit exactly when $\hat{\psi} = 1$. The distribution statement is Theorem 34.5.5 for one degree of freedom.

■

Part (c) says that grouping records with identical regressors and carrying the group size as an exposure is *exactly* a bookkeeping operation. It is a property of the Poisson, not of generalized linear models in general, and it comes from sums of independent Poissons being Poisson. The use is both practical — twenty thousand records collapse to a few dozen — and theoretical: the grouped cells may have large counts, so the deviance of a single fitted model is calibrated by Theorem 34.5.5 even though it is not for the individual records.

Example. Twenty thousand records in thirty-two rows

Take the physician-visit data of Example (Physician visits in a health insurance experiment) and keep only categorical regressors: the individual-deductible indicator, the three self-rated health indicators, and the chronic-disease score banded into four groups. There are 32 distinct covariate patterns, holding between 9 and 3832 person-years. Collapse the 20190 records to these rows, recording the total visits and the number of person-years in each, and fit the rate model with offset the log of the person-years. The coefficients agree with the person-level fit to fourteen decimal places, as [Proposition 37.2.1\(c\)](#) promises.

Read as rates, a person-year in excellent health, in the lowest disease band and on a plan without an individual deductible, is expected to produce 1.9758 visits; an individual deductible multiplies that by 0.8247, poor self-rated health by 1.6875, and the highest disease band by 2.2621.

The grouped deviance is 665.60 on 24 degrees of freedom, and now the χ^2 calibration applies, since the smallest fitted cell mean is 24.7. The model is rejected. Where does the misfit

sit? The fitted model has main effects only, so its 24 residual degrees of freedom are the interactions among the three factors; adding every two-factor interaction among them, 23 parameters in all, drops the deviance to 106.87 on 9 degrees of freedom, a fall of 558.7 on 15. Five sixths of the misfit is therefore mean model, not variance function. What survives the repair — 106.87 on nine degrees of freedom, still far beyond any plausible χ^2 — is the Poisson variance function, which [Section 37.4](#) takes up.

```
import numpy as np
import pandas as pd
import statsmodels.api as sm

def scoring(X, y, offset=0.0, tol=1e-11, maxit=60):
    """Fisher scoring for a Poisson log-linear model with a fixed offset."""
    beta = np.zeros(X.shape[1])
    beta[0] = np.log(max(y.mean(), 1e-6))
    for _ in range(maxit):
        mu = np.exp(X @ beta + offset)
        z = X @ beta + (y - mu) / mu           # working response
        XW = X * mu[:, None]                  # weights w_i = mu_i
        step = np.linalg.solve(X.T @ XW, XW.T @ z)
        if np.max(np.abs(step - beta)) < tol:
            return step
        beta = step
    return beta

data = sm.datasets.randhie.load_pandas().data.copy()
data["band"] = pd.cut(data["disea"], [-0.1, 6, 10, 15, 60], labels=[0, 1, 2, 3])
cells = ["idp", "hlthg", "hlthf", "hlthp", "band"] # all categorical
y = data["mdvis"].to_numpy(float)

def design(frame):
    """Intercept, the plan and health indicators, and three disease-band indicators.
    """
    columns = [np.ones(len(frame))] + [frame[v].to_numpy(float)
                                         for v in ["idp", "hlthg", "hlthf", "hlthp"]]
    band = frame["band"].to_numpy(int)
    return np.column_stack(columns + [(band == k).astype(float) for k in (1, 2, 3)])

grouped = data.groupby(cells, as_index=False, observed=True).agg(
    visits=("mdvis", "sum"), person_years=("mdvis", "size"))
s = grouped["visits"].to_numpy(float)           # total visits in the group
t = grouped["person_years"].to_numpy(float)      # exposure: person-years
X, Xg = design(data), design(grouped)

beta_person = scoring(X, y)                     # one row per person-year
beta_group = scoring(Xg, s, offset=np.log(t))    # one row per pattern, offset
log t

print("groups:", len(s), " largest coefficient difference:",
      np.abs(beta_person - beta_group).max())
print("fitted rate, free-plan person-years in excellent health:",
      np.exp(beta_group[0]))
```

[Run this code online](#)

37.2.2 What a free exposure coefficient buys and costs

Why fix $\psi = 1$ instead of estimating it, as [Proposition 37.2.1\(d\)](#) allows? It depends on where the exposure came from.

When exposure is a bookkeeping device — the grouping of part (c), where the individual records are the real data — proportionality is a theorem, not an assumption, and a $\hat{\psi}$ far from one is a symptom of misspecification elsewhere, because $\log n_g$ is free to proxy for anything correlated with group size. In [Example \(Twenty thousand records in thirty-two rows\)](#) it is $\hat{\psi} = 1.2266$, standard error 0.0146, interval (1.1980, 1.2553), with likelihood ratio statistic 249.019 against $\psi = 1$. A person-year cannot contribute more than proportionally to its own visits; the statistic is detecting the same lack of fit as the deviance of 665.60.

When exposure genuinely varies across units — different follow-up times, different populations — proportionality is a substantive assumption, and there is a real trade-off, which a simulation makes concrete.

Example. Three ways to handle an exposure

Counts are generated for $m = 300$ units from $\log \mu_i = 0.4 + 0.5x_i + \psi \log t_i$, with x_i standard normal and the exposure t_i correlated with it. Three analyses are run on each of 2000 data sets — the correct offset, a free coefficient on $\log t$, and no exposure term — and [Figure 37.2.1](#) shows the sampling distributions of the estimated slope of x , whose true value is 0.5.

When the truth is $\psi = 1$, both the offset and the free-coefficient model are unbiased, with means 0.500 and 0.500, but the free coefficient has standard deviation 0.029 against 0.017, a loss of precision by a factor of 1.772: estimating a parameter you already know costs information. Dropping the exposure is a disaster, the slope averaging 1.096, because x is left to explain the variation in exposure as well as in the rate.

When the truth is $\psi = 0.7$, the free-coefficient model is still centred at 0.501, while the offset model, forcing a wrong proportionality, is centred at 0.322 — a bias of more than six of its own standard deviations. Fixing a coefficient you have got wrong costs more than estimating one you knew.

Warning. The third analysis fails by ordinary omitted variable bias: $\log t$ belongs in the linear predictor, and leaving it out loads its effect onto whatever is correlated with it. What is special about an offset is only that its coefficient is known, so it is not a variable to be selected and must never be dropped by a stepwise procedure or a lasso.

37.2.3 Rates elsewhere

Two further uses of the same device are worth naming.

Piecewise-constant hazards. Split each subject's follow-up into intervals on which the hazard is constant, and let d_{ik} be subject i 's events in interval k and t_{ik} the time at risk. Treating the d_{ik} as independent Poisson counts with means $t_{ik} \exp(\alpha_k + \mathbf{x}_{(i)}^\top \boldsymbol{\beta})$ gives a proportional-hazards

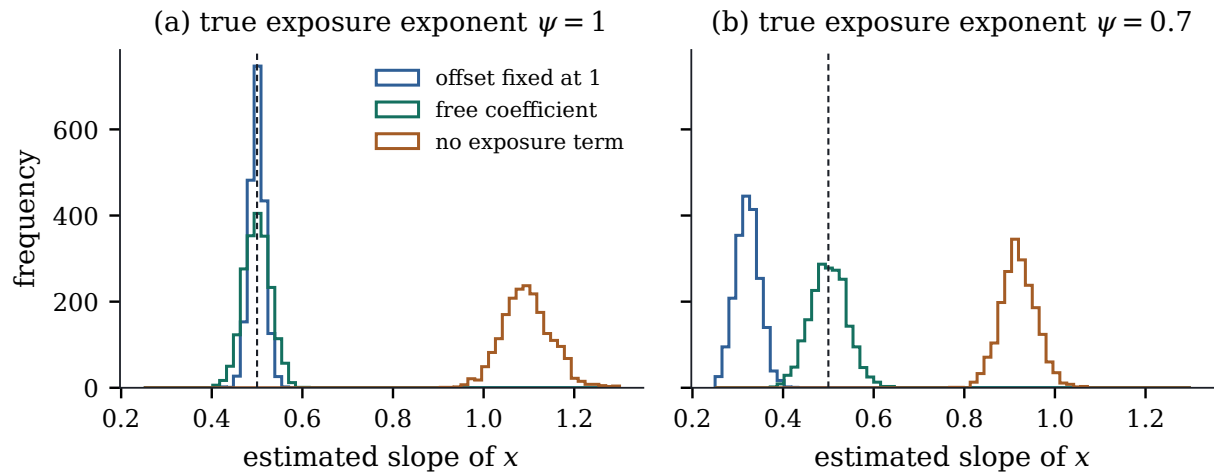


Figure 37.2.1: Sampling distributions of the estimated slope of x over 2000 simulated data sets, under three treatments of the exposure, when the true exposure exponent is (a) one and (b) 0.7. The dashed line marks the true slope.

model with a step-function baseline, fitted as an ordinary Poisson regression with offset $\log t_{ik}$ and one parameter per interval ([Exercise \(B1\)](#)); the connection is due to Holford and to Laird and Olivier.

Standardization. A standardized morbidity ratio divides the observed count by an expected count E_i from external reference rates; taking $\log E_i$ as the offset makes $\exp(\mathbf{x}_{(i)}^\top \boldsymbol{\beta})$ the ratio itself.

Fixing the exposure coefficient is right when theory or bookkeeping says the count is proportional to the exposure, and costs nothing; estimating it is right when proportionality is in doubt, and costs precision. What is never right is leaving the exposure out.

37.2.4 Exercises

37.2.4.1 A. Check your understanding

Exercise. A1

A rate model uses person-years as the exposure. A colleague re-expresses exposure in person-months, multiplying every t_i by twelve. Which coefficients change and by how much? Which fitted values change?

Exercise. A2

Explain why regressing $\log(y_i/t_i)$ on $\mathbf{x}_{(i)}$ by least squares is not equivalent to fitting [Eq. 37.2.1](#), giving two distinct reasons.

37.2.4.2 B. Practice

Exercise. B1

Suppose subject i is followed until an event or censoring, with a hazard that is constant at $h_{ik} = \exp(\alpha_k + \mathbf{x}_{(i)}^\top \boldsymbol{\beta})$ on interval k . Write $d_{ik} \in \{0, 1\}$ for the event indicator and t_{ik}

for the time at risk in that interval. Show that the likelihood contribution of subject i is $\prod_k h_{ik}^{d_{ik}} e^{-h_{ik}t_{ik}}$, and hence that maximizing it is the same as fitting a Poisson log-linear model to the d_{ik} with offset $\log t_{ik}$.

Solution. With a constant hazard h on an interval, the probability of surviving a time t is e^{-ht} and the density of an event at the end is he^{-ht} , so the contribution of interval k is $h_{ik}^{d_{ik}} e^{-h_{ik}t_{ik}}$, and the contributions multiply. Its logarithm is $\sum_{i,k} \{d_{ik} \log h_{ik} - h_{ik}t_{ik}\}$. The Poisson log-likelihood for counts d_{ik} with means $t_{ik}h_{ik}$ differs from it by $\sum_{i,k} d_{ik} \log t_{ik}$, a constant in the parameters, so the maximizers agree. The d_{ik} need not actually be Poisson; only the likelihoods coincide.

Exercise. B2

In the setting of [Example \(Three ways to handle an exposure\)](#), suppose the analyst tests $H_0 : \psi = 1$ and uses the offset model if the test does not reject. Explain why the resulting estimator of the slope of x has a sampling distribution that is neither of the two shown in [Figure 37.2.1](#), and relate the phenomenon to the pre-test problem [Example \(Coverage after a pretest\)](#).

37.2.4.3 C. Going deeper

Exercise. C1

Extend [Proposition 37.2.1\(c\)](#) to the case where the individual records already carry exposures t_i that differ within a group. Show that the grouped fit with exposure $\sum_{i \in g} t_i$ reproduces the individual fit, and identify exactly what has to be constant within a group.

Solution. What must be constant within a group is the row $\mathbf{x}_{(i)}$, not the exposure. If $\mathbf{x}_{(i)} = \mathbf{x}_{(g)}$ for all $i \in g$ then $\mu_i = t_i \mu_g^*$ with $\mu_g^* = e^{\mathbf{x}_{(g)}^\top \boldsymbol{\beta}}$, so $\sum_{i \in g} \mu_i = (\sum_{i \in g} t_i) \mu_g^*$ and the log-likelihood is $\sum_g \{s_g \log \mu_g^* - (\sum_{i \in g} t_i) \mu_g^*\} + \sum_i y_i \log t_i$, whose last term is free of $\boldsymbol{\beta}$. Sums of independent Poissons are Poisson, so the grouped data have exactly the claimed distribution.

Exercise. C2

Let the truth be $\log \mu_i = \psi \log t_i + \beta_0 + \beta_1 x_i$ with $\psi \neq 1$, and suppose the offset model is fitted. Show that the limiting value of $\hat{\beta}_1$ solves $E\{\mathbf{x} t (t^{\psi-1} e^{\beta_0 + \beta_1 x} - e^{\beta_0^* + \beta_1^* x})\} = \mathbf{0}$, and argue that $\beta_1^* = \beta_1$ when $t^{\psi-1}$ is independent of x under the weighting by t .